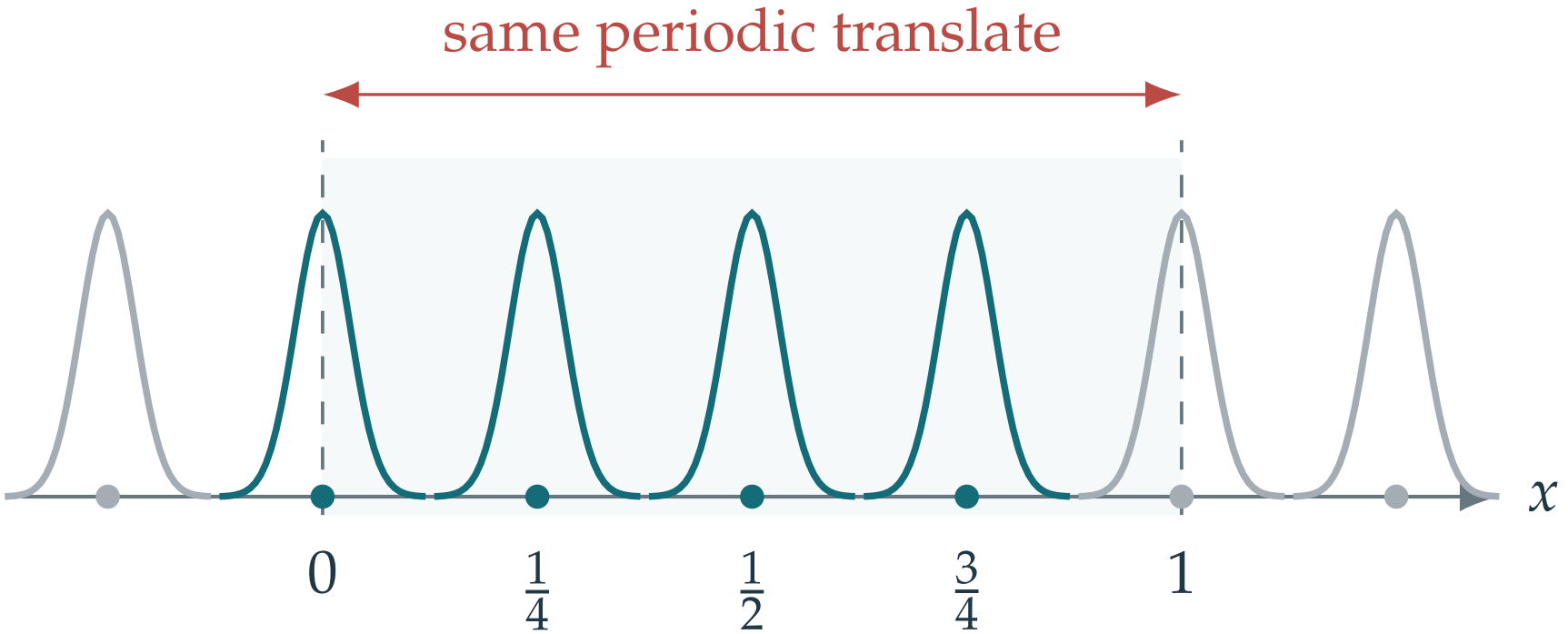

$$j = -2: \quad 2^j n \in \{0, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}\}$$

$$0 \le n < 2^{-j}, \qquad n \sim n + 2^{-j}$$